

Trigonometric functions



Outline

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Trigonometric functions



Introduction: Trigonometric functions

- ▶ In this section we introduce the **sine** and **cosine** functions and, from them, the **tangent**, **cotangent**, **secant** and **cosecant**
- ▶ We study their symmetry, periodicity, sign and monotonicity, with the help of their graphs and of the unit circle
- ▶ Trigonometric functions are the natural tool to describe periodic phenomena and oscillations

Sine and cosine

Definition 1: sine function and cosine function

The trigonometric functions:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \sin x$$

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \cos x$$

*are called **sine** and **cosine**.*

Symmetry and periodicity

- ▶ The sine is an **odd** function, while the cosine is an **even** function. Hence we have:

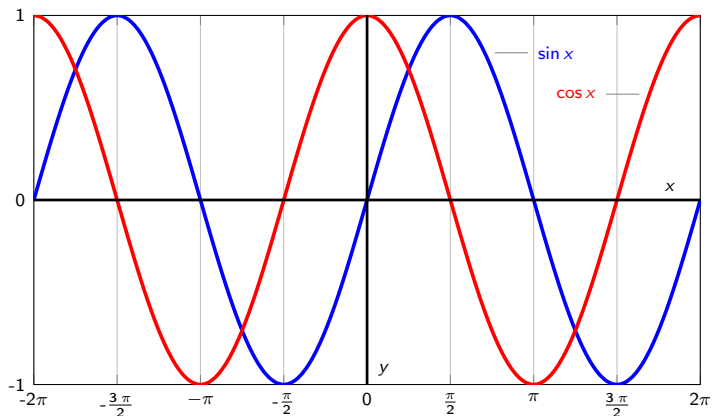
$$\sin(-x) = -\sin x, \quad \forall x \in \mathbb{R}$$

$$\cos(-x) = \cos x, \quad \forall x \in \mathbb{R}$$

- ▶ The sine and the cosine are periodic functions with period $T = 2\pi$ and we have the fundamental identity

$$\sin^2 x + \cos^2 x = 1, \quad \forall x \in \mathbb{R}$$

Graphs of sine and cosine



We have

$$\cos\left(x - \frac{\pi}{2}\right) = \sin x, \quad \forall x \in \mathbb{R}$$

Sign and monotonicity of the sine

- From the sign and the monotonicity of the sine function we have:

$$\sin x = 0 \iff x = k \pi, \forall k \in \mathbb{Z}$$

$$\begin{cases} \sin x > 0 & \text{if } x \in (2k\pi, 2k\pi + \pi), \forall k \in \mathbb{Z} \\ \sin x < 0 & \text{if } x \in (2k\pi - \pi, 2k\pi), \forall k \in \mathbb{Z} \end{cases}$$

$$\begin{cases} \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

Sign and monotonicity of the cosine

- From the sign and the monotonicity of the cosine function we have:

$$\cos x = 0 \iff x = k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}$$

$$\begin{cases} \cos x > 0 & \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} \\ \cos x < 0 & \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} \end{cases}$$

$$\begin{cases} \text{if } x \in (2k\pi + \pi, 2k\pi + 2\pi), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi, 2k\pi + \pi), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

Tangent and cotangent

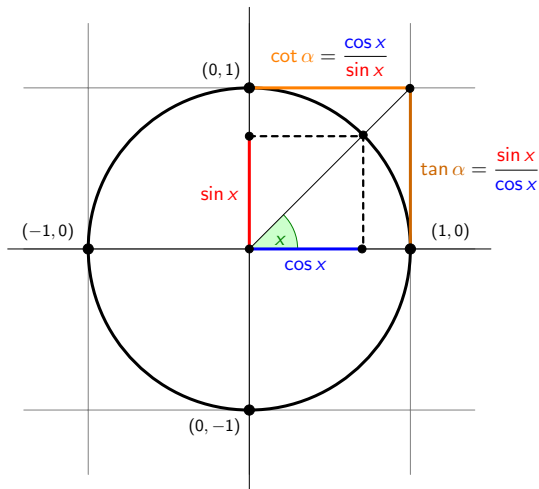
The trigonometric functions:

$$f : \mathbb{R} \setminus \left\{ k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z} \right\} \rightarrow \mathbb{R}, \quad f : x \mapsto \tan x = \frac{\sin x}{\cos x}$$

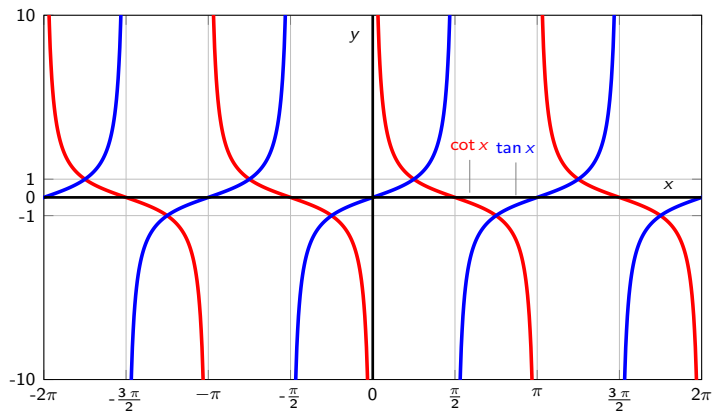
$$f : \mathbb{R} \setminus \{ k\pi, \forall k \in \mathbb{Z} \} \rightarrow \mathbb{R}, \quad f : x \mapsto \cot x = \frac{\cos x}{\sin x}$$

are called **tangent** and **cotangent**; they are both odd and periodic with period $T = \pi$.

The unit circle



Graphs of tangent and cotangent



Secant and cosecant

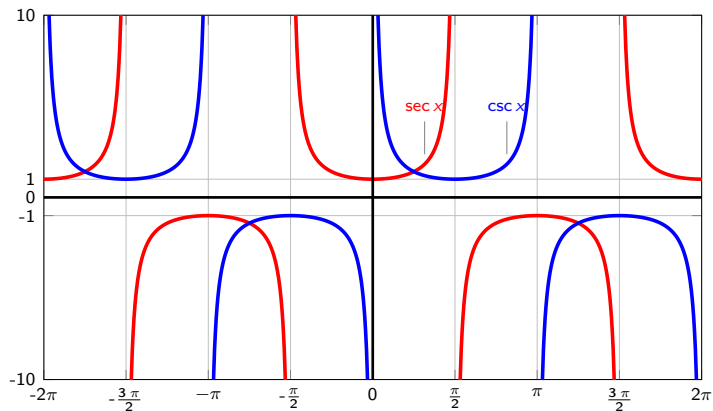
The trigonometric functions:

$$f : \mathbb{R} \setminus \{k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f : x \mapsto \sec x = \frac{1}{\cos x}$$

$$f : \mathbb{R} \setminus \{k\pi, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f : x \mapsto \csc x = \frac{1}{\sin x}$$

are called **secant** and **cosecant** and they are periodic with period $T = 2\pi$. The secant function is even, while the cosecant function is odd.

Graphs of secant and cosecant



Summary

Trigonometric functions:

- ▶ $\sin$ (odd) and $\cos$ (even) are defined on $\mathbb{R}$, periodic with period 2π , with $\sin^2 x + \cos^2 x = 1$
- ▶ $\tan = \frac{\sin}{\cos}$ and $\cot = \frac{\cos}{\sin}$: odd, periodic with period π
- ▶ $\sec = \frac{1}{\cos}$ (even) and $\csc = \frac{1}{\sin}$ (odd): periodic with period 2π
- ▶ Sign and monotonicity can be read on the unit circle

Values of the trigonometric functions



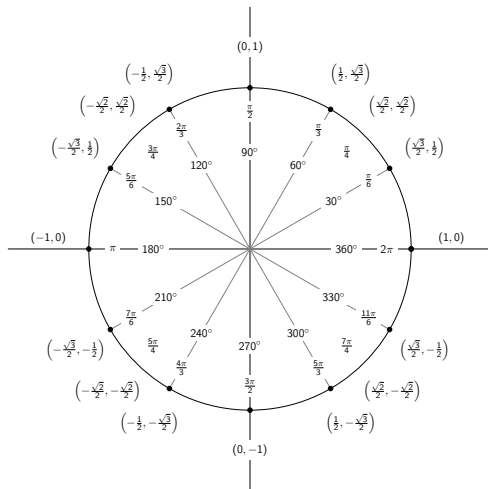
Introduction: Values of the trigonometric functions

- ▶ In this section we collect the **notable values** of the trigonometric functions at the most common angles
- ▶ We also see how the functions change for **related angles** ($-x$, $\frac{\pi}{2} \pm x$, $\pi \pm x$)
- ▶ These tables are used constantly in exercises

Notable values

	cos	sin	tan	cot	sec	csc
0	1	0	0	$\pm\infty$	1	$\pm\infty$
$x = \frac{\pi}{6} (30^\circ)$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$	$\frac{2}{3}\sqrt{3}$	2
$x = \frac{\pi}{4} (45^\circ)$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1	$\sqrt{2}$	$\sqrt{2}$
$x = \frac{\pi}{3} (60^\circ)$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2}{3}\sqrt{3}$
$x = \frac{\pi}{2} (90^\circ)$	0	1	$\pm\infty$	0	$\pm\infty$	1
$x = \pi (180^\circ)$	-1	0	0	$\pm\infty$	-1	$\pm\infty$

Notable values on the unit circle



Related angles

	cos	sin	tan	cot
x	$\cos x$	$\sin x$	$\tan x$	$\cot x$
$-x$	$\cos x$	$-\sin x$	$-\tan x$	$-\cot x$
$\frac{\pi}{2} + x$	$-\sin x$	$\cos x$	$-\cot x$	$-\tan x$
$\frac{\pi}{2} - x$	$\sin x$	$\cos x$	$\cot x$	$\tan x$
$\pi + x$	$-\cos x$	$-\sin x$	$\tan x$	$\cot x$
$\pi - x$	$-\cos x$	$\sin x$	$-\tan x$	$-\cot x$

Summary

Values of the trigonometric functions:

- ▶ Notable values at $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \pi$ of $\cos, \sin, \tan, \cot, \sec, \csc$
- ▶ Related angles: the functions at $-x, \frac{\pi}{2} \pm x, \pi \pm x$ are expressed through the functions at x

Main trigonometric formulas

Introduction: Main trigonometric formulas

- ▶ In this section we collect the main **trigonometric identities**
- ▶ Relations among the functions, addition, double-angle, triple-angle, half-angle, parametric, sum-to-product and product-to-sum formulas
- ▶ They are essential to simplify expressions, solve equations and compute limits and integrals

Relations among the trigonometric functions

	$\sin x$	$\cos x$	$\tan x$	$\cot x$
$\sin x$	-	$\pm\sqrt{1 - \cos^2 x}$	$\pm\sqrt{\frac{\tan^2 x}{1 + \tan^2 x}}$	$\pm\sqrt{\frac{1}{1 + \cot^2 x}}$
$\cos x$	$\pm\sqrt{1 - \sin^2 x}$	-	$\pm\sqrt{\frac{1}{1 + \tan^2 x}}$	$\pm\sqrt{\frac{\cot^2 x}{1 + \cot^2 x}}$
$\tan x$	$\pm\sqrt{\frac{\sin^2 x}{1 - \sin^2 x}}$	$\pm\sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$	-	$\frac{1}{\cot x}$
$\cot x$	$\pm\sqrt{\frac{1 - \sin^2 x}{\sin^2 x}}$	$\pm\sqrt{\frac{\cos^2 x}{1 - \cos^2 x}}$	$\frac{1}{\tan x}$	-

where the symbol $\pm$ means that the sign depends on the quadrant in which x lies.

Addition and subtraction

$$\sin(x_1 \pm x_2) = \sin x_1 \cos x_2 \pm \sin x_2 \cos x_1$$

$$\cos(x_1 \pm x_2) = \cos x_1 \cos x_2 \mp \sin x_1 \sin x_2$$

$$\tan(x_1 \pm x_2) = \frac{\tan x_1 \pm \tan x_2}{1 \mp \tan x_1 \tan x_2}$$

$$\cot(x_1 \pm x_2) = \frac{\cot x_1 \cot x_2 \mp 1}{\cot x_2 \pm \cot x_1}$$

Double-angle formulas

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2 \cos^2 x - 1$$

$$= 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\cot(2x) = \frac{\cot^2 x - 1}{2 \cot x}$$

Triple-angle formulas

$$\sin(3x) = 3 \sin x - 4 \sin^3 x$$

$$\cos(3x) = 4 \cos^3 x - 3 \cos x$$

Half-angle formulas (1/2)

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}$$

$$\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$$

Half-angle formulas (2/2)

$$\tan \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$

$$\cot \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{1 - \cos x}} = \frac{\sin x}{1 - \cos x} = \frac{1 + \cos x}{\sin x}$$

Parametric formulas (tangent half-angle substitution)

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

Sum-to-product formulas (prosthaphaeresis)

$$\sin x_1 + \sin x_2 = 2 \sin \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2}$$

$$\sin x_1 - \sin x_2 = 2 \sin \frac{x_1 - x_2}{2} \cos \frac{x_1 + x_2}{2}$$

$$\cos x_1 + \cos x_2 = 2 \cos \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2}$$

$$\cos x_1 - \cos x_2 = -2 \sin \frac{x_1 + x_2}{2} \sin \frac{x_1 - x_2}{2}$$

Product-to-sum formulas (Werner's formulas)

$$\sin x_1 \sin x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) - \cos(x_1 + x_2) \right)$$

$$\sin x_1 \cos x_2 = \frac{1}{2} \left(\sin(x_1 - x_2) + \sin(x_1 + x_2) \right)$$

$$\cos x_1 \cos x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) + \cos(x_1 + x_2) \right)$$

Additional formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\sin x \cos x = \frac{\sin(2x)}{2}$$

Summary

Main trigonometric formulas:

- ▶ Each trigonometric function can be expressed through the others (up to the sign, which depends on the quadrant)
- ▶ **Addition/subtraction, double-angle, triple-angle, half-angle** formulas
- ▶ **Parametric** formulas: $\sin x$ and $\cos x$ in terms of $\tan \frac{x}{2}$
- ▶ **Sum-to-product** (prosthaphaeresis) and **product-to-sum** (Werner) formulas
- ▶ $\sin^2 x$, $\cos^2 x$ and $\sin x \cos x$ in terms of the double angle

Final summary

- ▶ **Sine** and **cosine**: periodic with period 2π , odd and even respectively, with $\sin^2 x + \cos^2 x = 1$
- ▶ **Tangent**, **cotangent** (period π), **secant** and **cosecant** (period 2π) are built from sine and cosine
- ▶ Tables of **notable values** and of **related angles**
- ▶ The **trigonometric formulas** allow us to transform and simplify expressions